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The Null Directions of the Tetrad Frame Carry a Complete Boolean Basis

Stephen Nelson

Field Dynamics Research
stephen@fielddynamics.org

Abstract
Requiring four null directions of a Lorentzian frame to be isotropic forces a regular tetrahedron; with its antipodes it gives eight null directions, the points of a stellated octahedron and the corners of a cube. Gravity Field Dynamics has claimed that this sampled null cone is a universal Boolean operator. We make the claim precise and prove the strong form. The signs of the eight directions are the eight three-bit words, so functions on them are the 256 three-input Boolean functions, and the Walsh–Hadamard matrix H_8 represents every one; that much holds for any eight labelled points. Gate universality is different. The Walsh characters are parities, and parity gates, including the TetA/TetB split, generate only the 16 affine functions, so by Post's theorem they are not universal. We show that $H_8 = S\,D\,S^T$, where S is the evaluation table of the AND monomials, which is its own inverse over GF(2) and gives every function a unique algebraic normal form, and $D = e^{(\ln 2 + i\pi)|x|}$ is a half-turn per bit. The Walsh read therefore factors into an AND table and an XOR table, which together are universal. Reading the two-bit block as spinor indices, the truth table of a two-input gate is a 4-vector whose Minkowski norm is its determinant. Exactly ten gates are null, the separable ones; they include AND and NOT, and NOR, a real light ray, is universal on its own. Parity lies off the cone.

Keywords: null cone, tetrad, stellated octahedron, Boolean functions, functional completeness, Walsh–Hadamard transform, algebraic normal form, null rotation

1 Introduction

Letter 1 of the tetrad series derives the stellated octahedron from the null cone [1]: four null directions that introduce no preferred spatial direction must form a regular tetrahedron on the celestial sphere, and with their antipodes they are the eight points of the star. The archive goes on to claim that this sampled null cone is a universal Boolean operator [2], on the strength of two facts: the eight corners of the cube carry every three-input Boolean function, and a coupled oscillator laid out on them, the octahedral resonance node, computes with the Walsh spectrum. The archive states the conjunction as a claim.

"Universal" has two meanings, and they come apart here. A set of points is universal in the weak sense if every Boolean function on it can be represented; any eight labelled points are. A set of gates is universal, or functionally complete, if composing them produces every Boolean function [4]. This paper proves the strong form for the null directions of the frame, and shows where in their algebra the universality sits. Figure 1 summarises the results.

2 The Eight Null Directions

Proposition 1. *In a Lorentzian frame, let $u_1, \dots, u_4$ be unit spatial vectors with zero centroid and isotropic second moment, so that each $k_I = (1, u_I)$ is null. Then the u_I are a regular tetrahedron, with $u_I \cdot u_J = -\frac{1}{3}$, and up to rotation they are $v/\sqrt{3}$ for the four corners $v \in \{\pm 1\}^3$ with $v_1 v_2 v_3 = +1$ (TetA). With their antipodes (TetB) they are the eight corners of the cube. Writing each corner's signs as bits, $b_i = (1 - v_i)/2$, the eight null directions are the eight words $b \in \{0,1\}^3$, and TetA and TetB are the words of even and odd weight.*

Proof. The tetrahedron is Letter 1's proposition [1]; squaring the centroid condition gives $4 + 12c = 0$. The rest is the substitution $v_i = (-1)^{b_i}$, under which $v_1 v_2 v_3 = (-1)^{|b|}$. □

So a function on the eight null directions is exactly a three-input Boolean function, and there are $2^8 = 256$ of them.

3 Representation Versus Gates

Let H_8 have entries $(-1)^{|i \wedge j|}$, the Walsh characters of the cube [8, 9], where $i \wedge j$ is the bitwise AND of the words and $|\cdot|$ counts ones.

Proposition 2 (representation). *$H_8^2 = 8I$, so every function on the eight null directions is a unique real combination of the eight Walsh characters.*

This is the sense of the archive's claim, and it is generic: the same holds for any eight labelled points. It says nothing about which gates the structure supplies. The Walsh characters themselves are, in 0/1 form, the parities $x_J = \bigoplus_{i \in J} b_i$.

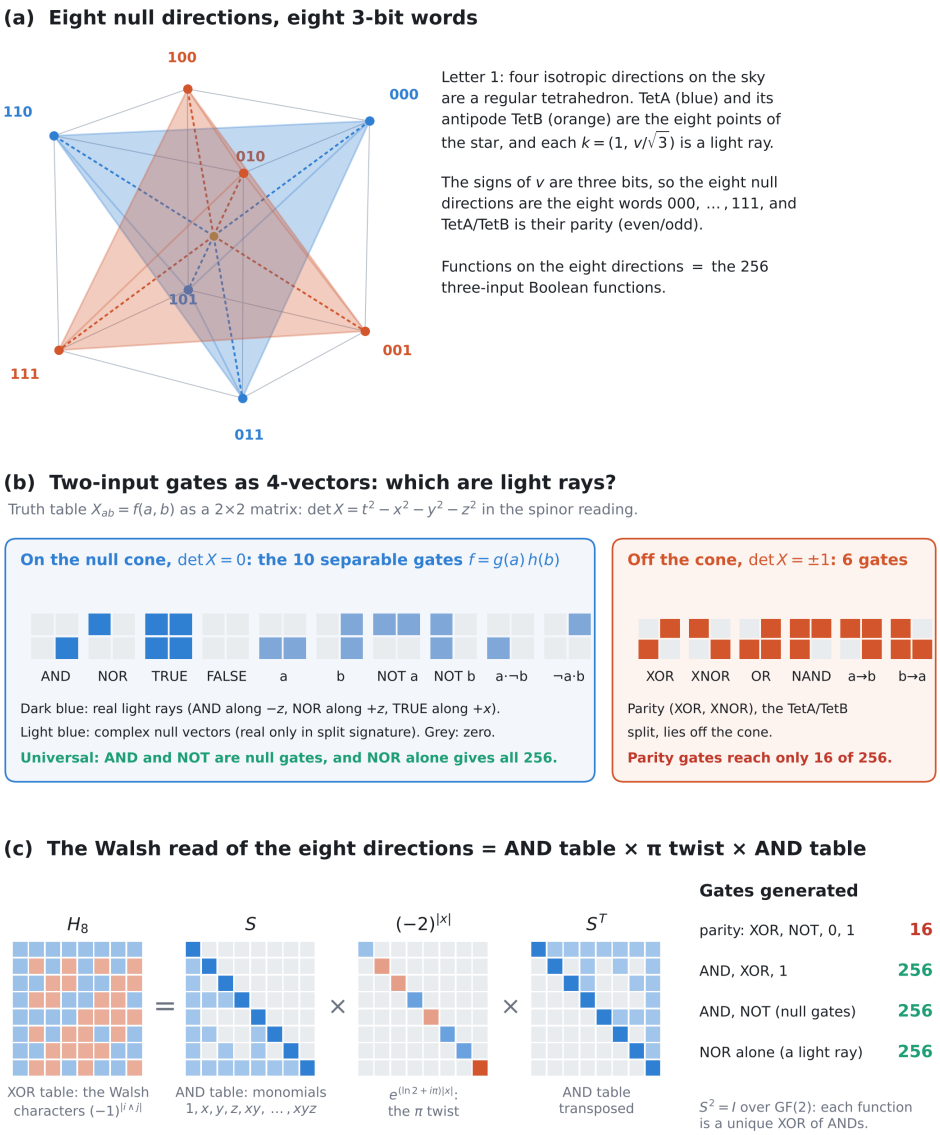


Figure 1. (a) The eight null directions of the tetrad frame (Letter 1), labelled by the signs of their spatial parts: the eight three-bit words, with TetA and TetB the even and odd words. (b) The 16 two-input gates, with truth tables read as 2×2 matrices X. In the spinor reading $\det X$ is the Minkowski norm. Ten gates are null, the separable gates $g(a)h(b)$; AND, NOR and TRUE are real light rays. The other six, including parity, are off the cone (Theorem 3). (c) The Walsh read of the eight directions factors as an AND table, a half-turn per bit, and the transposed AND table (Theorem 2). Right: the number of three-input functions each gate set generates.

Theorem 1 (parity is not universal). *Every Walsh character, and the TetA/TetB indicator, is affine over GF(2): its algebraic normal form has degree at most one. Closing the three projections under XOR, NOT and the constants gives exactly the* 16 *affine functions of three bits, not all* 256*. So no set of parity gates is universal.*

Proof. A parity is a sum of variables over GF(2), and compositions of affine functions are affine. The count of 16 is checked by exhaustive closure. By Post's theorem [4] a gate set inside the affine class is not functionally complete. □

In particular the XOR read off the octahedral node's grade-two Walsh coefficient, and the parity term it reads at grade three, are affine. The node computes NAND and the full adder because its readout is a real linear combination of Walsh coefficients, which is Proposition 2, not because parity gates are universal.

4 The Walsh Read Factors Into AND and XOR

Let S be the 8×8 matrix with $S_{ij} = 1$ when the bits of j are a subset of those of i, and 0 otherwise. Equivalently, $S = A^{\otimes 3}$ with $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, the Sierpiński triangle.

Theorem 2. *(i) S_{ij} is the AND of the bits of j at the corner i: S is the evaluation table of the eight AND monomials* $1, x, y, z, xy, xz, yz, xyz$.

(ii) $S^2 = I$ *over GF(2), so every Boolean function is a unique XOR of AND monomials, its algebraic normal form, with coefficients* $Sf \bmod 2$. *(iii)*

$$H_8 = S\,D\,S^T, \qquad D = \operatorname{diag}\big((-2)^{|x|}\big) = e^{(\ln 2 + i\pi)|x|} \tag{1}$$

(iv) AND, XOR and the constant 1 *generate all* 256 *functions.*

Proof. (i) A monomial $\prod_{b\in J} x_b$ is 1 at a corner exactly when every bit of J is set there. (ii) $A^2 \equiv I \pmod 2$, and tensor powers follow. This is the Zhegalkin, or Reed–Muller, expansion [5, 6]. (iii) Entry by entry, with $n = |i \wedge j|$, $\sum_{K\subseteq i\wedge j}(-2)^{|K|} = (1-2)^n = (-1)^n$. (iv) follows from (ii), and is also checked by exhaustive closure. □

So the Walsh read of the eight null directions factors into the two halves of a universal gate set. The outer factors are the AND table; the half-turn $e^{i\pi}$ per bit in the middle carries the parity signs. The universality of the structure is located in the Sierpiński factor, not in the Hadamard sign pattern. In the spinor reading of the next section, A acting on both indices is a null rotation [3].

5 The Null Cone of the Two-Bit Block

Write a two-input gate as its truth table $X_{ab} = f(a, b)$, a 2×2 matrix, and read the two bits as the two spinor indices of a 4-vector, $X = t + x\sigma_1 + y\sigma_2 + z\sigma_3$, so that $\det X = t^2 - x^2 - y^2 - z^2$ [10]. This is the reading in which the tetrad frame's legs form a Newman–Penrose null tetrad [3].

Theorem 3 (the null gates). *A gate is null,* $\det X = 0$, *exactly when* X *has rank at most one, that is, when* $f(a, b) = g(a)\,h(b)$. *Exactly* 10 *of the* 16 *gates are null: FALSE, TRUE, AND, NOR, the four literals* a, b, $\neg a$, $\neg b$, *and* $a \wedge \neg b$, $\neg a \wedge b$. *The other six, XOR, XNOR, OR, NAND and the two implications, have* $\det X = \pm 1$. *Among the null gates, AND, NOR and TRUE are real light rays, along* $-z$, $+z$ *and* $+x$; *FALSE is the zero vector; the remaining six are complex null vectors, real only in the split-signature reading of real* 2×2 *matrices. AND is the light ray* $|11\rangle$ *fixed by the Sierpiński null rotation* $X \mapsto AXA^T$.

Proof. A 2×2 matrix is singular exactly when its rank is at most one, and a 0/1 matrix of rank one is an outer product $g\,h^T$ of 0/1 vectors. The components follow from $x^\mu = \frac{1}{2}\mathrm{Tr}(\sigma_\mu X)$; a non-symmetric X has an imaginary y component. A fixes $|1\rangle$. □

Corollary. *The null gates are a universal set: they contain AND and NOT. NOR, a real light ray, is universal on its own. Parity, XOR and XNOR, lies off the cone.*

Proof. {AND, NOT} and {NOR} are classical complete sets [4, 7]; exhaustive closure confirms that each generates all 256 three-input functions. □

Rank-one truth tables are the combinatorial rectangles of communication complexity [11]; the observation here is that, in the spinor reading of the frame, they are its light rays.

6 Discussion and Scope

The archive's claim, that the null cone of the frame is a universal Boolean operator, is true in two senses, and they should be kept apart. In the weak sense, representation, it holds for the eight null directions as for any eight labelled points (Proposition 2). In the strong sense, functional completeness, it holds in a specific way. Parity alone, the structure that separates TetA from TetB, is not universal (Theorem 1). The Walsh read of the eight directions factors into an AND table and a half-turn (Theorem 2), and on the two-bit block the light rays are exactly the separable gates, which are universal, while parity is off the cone (Theorem 3).

What is standard and what is new: Post's theorem, the algebraic normal form, the completeness of NOR, and the spinor–vector correspondence are classical. The new content is their placement on the frame: the null directions as the three-bit words, the factorisation of their Walsh read into AND and XOR tables, and the identification of the separable gates with the light rays of the spinor reading.

The paper does not show that physical light computes, nor that a lattice of photons is a computing medium; those remain the archive's interpretation. The real form matters: only AND, NOR and TRUE are real light rays in Minkowski signature, and the other null gates are real only in split signature. The ten null and six non-null gates should not be confused with the archive's split of 16 into 10 symmetric and 6 antisymmetric components: the null gates do not form a linear subspace, and no relation between the two counts is claimed.

Every statement is checked by the accompanying script (18 checks, exact arithmetic, closures exhaustive over all 256 functions).

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